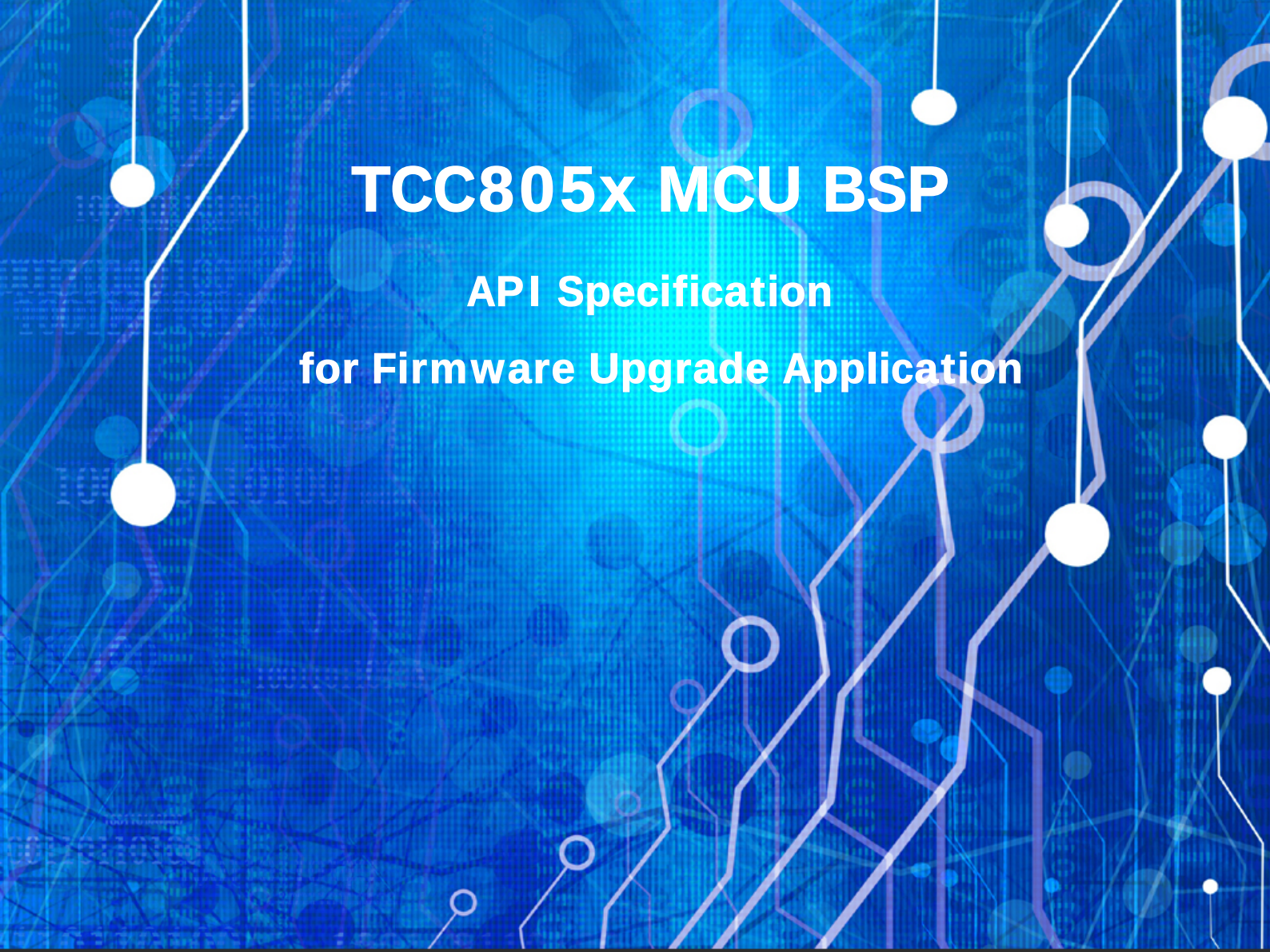




TCC805x MCU BSP

API Specification

for Firmware Upgrade Application

Rev. 1.01 [A]

2021-01-22

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1 INTRODUCTION

This document describes the Firmware upgrade sample application of TCC805x MCU BSP. Main purpose of this application is to upgrade R5 main F/W.

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

A72	Cortex-A72 multi core processor
R5	Cortex-R5 single core processor
FWUG	Firmware Upgrade
F/W	Firmware
S-NOR	Serial NOR flash memory
SRAM1	Internal SRAM1
MBOX	Mailbox
App	Application
tcc_snor_updater driver	Device driver on A72 for upgrading F/W of S-NOR
R5 sub F/W	R5 F/W binary image to upgrade R5 MCU BSP main F/W binary image (r5_sub_fw.rom)
R5 main F/W	R5 MCU BSP main F/W binary image (r5_fw_TCC805x.rom)
R5-BL1	Boot Loader 1 for R5 boot

3 SOURCE FILES

3.1 Location of Source Files and Project

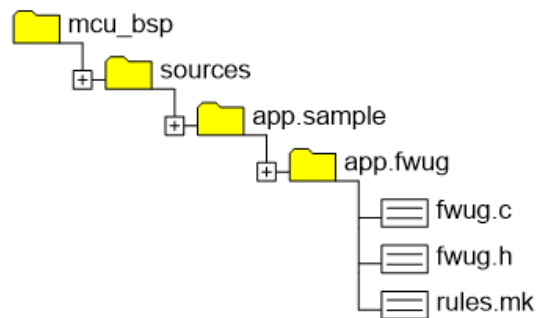


Figure 3.1 Location of FWUG Application Source Files

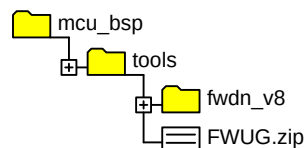


Figure 3.2 Location of R5 Sub F/W Project

3.2 Simple Description of Source Files and Project

- fwug.c
 - FWUG sample application source
- fwug.h
 - FWUG sample application header
- rules.mk
 - Rules used for build
- FWUG.zip
 - R5 sub F/W project

4 CONSTANTS

4.1 Base Address and Offset

Constant values for TCC805x memory base address

Declaration

```
#define FWUG_SNOR_BASEADDR    (0x08000000UL)
#define FWUG_SRAM1_BASEADDR   (0x07000000UL)
```

Description

Value	Description
FWUG_SNOR_BASEADDR	S-NOR base address
FWUG_SRAM1_BASEADDR	SRAM1 base address

Remark

None

4.2 Task Stack Size

Constant values for FWUG_Task stack size

Declaration

```
#define FWUG_TASK_STK_SIZE    (2048)
```

Description

Value	Description
FWUG_TASK_STK_SIZE	Stack size for FWUG_Task

Remark

None

4.3 MBOX Command

Constant values for MBOX command of FWUG

Declaration

```
#define FWUG_CMD_UPDATE_START    (0x01)
```

Description

Value	Description
FWUG_CMD_UPDATE_START	MBOX command to start F/W upgrade process. A72 sends this command to R5.

Remark

None

5 STRUCTURES

5.1 FWUGDev_T

Structure variables for MBOX device that FWUG uses

Declaration

```
typedef struct FWUGDev
{
    MBOXDvice_t *    dDev;
} FWUGDev_T;
```

Description

Value	Description
dDev	MBOX device description for FWUG process

Remark

None

5.2 FWUGMsg_T

Structure variables for MBOX messages that FWUG uses

Declaration

```
typedef struct FWUGMsg
{
    uint32    mCmd[MBOX_CMD_FIFO_SIZE];
    uint32    mData[MBOX_DATA_FIFO_SIZE];
} FWUGMsg_T;
```

Description

Value	Description
mCmd	MBOX command
mData	MBOX data

Remark

None

6 APIs

6.1 FWUG_CreateApp

Creates FWUG_Task.

Declaration

```
void FWUG_CreateApp  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.2 FWUG_Task

The task for waiting a start command of FWUG process

Declaration

```
static void FWUG_Task  
(  
    void *    pArg  
);
```

Parameters

<i>Value</i>	<i>Description</i>
<i>pArg</i>	<i>[in] Argument that is passed to Task</i>

Return

None

Remark

None

6.3 FWUG_Start

Executes FWUG process if the received MBOX command is FWUG_CMD_UPDATE_START.

Declaration

```
static void FWUG_Start  
(  
    uint32    uiCmd  
);
```

Parameters

Value	Description
<i>uiCmd</i>	<i>[in] MBOX command</i>

Return

None

Remark

None

6.4 FWUG_CopyAndJump

Copies and executes R5 sub F/W.

Declaration

```
static void FWUG_CopyAndJump  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7 HOW TO USE APPLICATION

Upgrading R5 main F/W needs FWUG application (on R5 main F/W) and R5 sub F/W.

7.1 Description of FWUG Process

FWUG application waits for FWUG process start command. If the command is received, application copies and executes R5 sub F/W by using *FWUG_Start()*.

```
static void FWUG_Task( void *pArg )
{
    int32  ret;
    uint32 uiCmd;
    int8   cDevName[ 7 ];

    ret    = -1;
    uiCmd  = 0;

    cDevName[ 0 ] = 'F';
    cDevName[ 1 ] = 'W';
    cDevName[ 2 ] = 'U';
    cDevName[ 3 ] = 'G';
    cDevName[ 4 ] = ( int8 ) 0x00;
    cDevName[ 5 ] = ( int8 ) 0x00;
    cDevName[ 6 ] = ( int8 ) 0x00;

    ( void ) pArg; //unused

    gFWUGDevHandle.dDev = MBOX_DevOpen( MBOX_CH_CA72MP_NONSECURE, cDevName, 0 );

    while( 1 )
    {
        ( void ) SAL_MemSet( gFWUGRecvData.mCmd, 0x00, MBOX_CMD_FIFO_SIZE );
        ( void ) SAL_MemSet( gFWUGRecvData.mData, 0x00, MBOX_DATA_FIFO_SIZE );

        ret = MBOX_DevRecv( gFWUGDevHandle.dDev, gFWUGRecvData.mCmd, gFWUGRecvData.mData );

        if( ret >= 0 )
        {
            uiCmd = gFWUGRecvData.mCmd[ 0 ];

            FWUG_Start( uiCmd );
        }
        else
        {
            ret = -1;
        }

        ( void ) SAL_TaskSleep( 100 );
    }
}
```

Refer to the block diagram of FWUG process below.

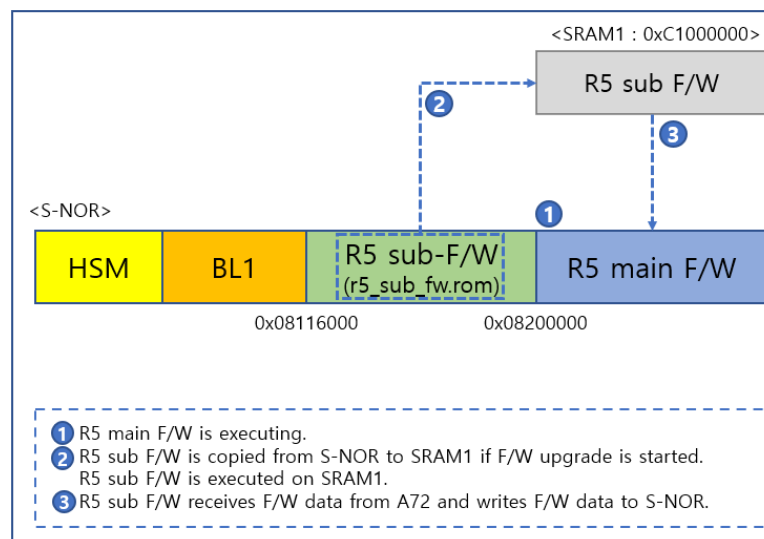
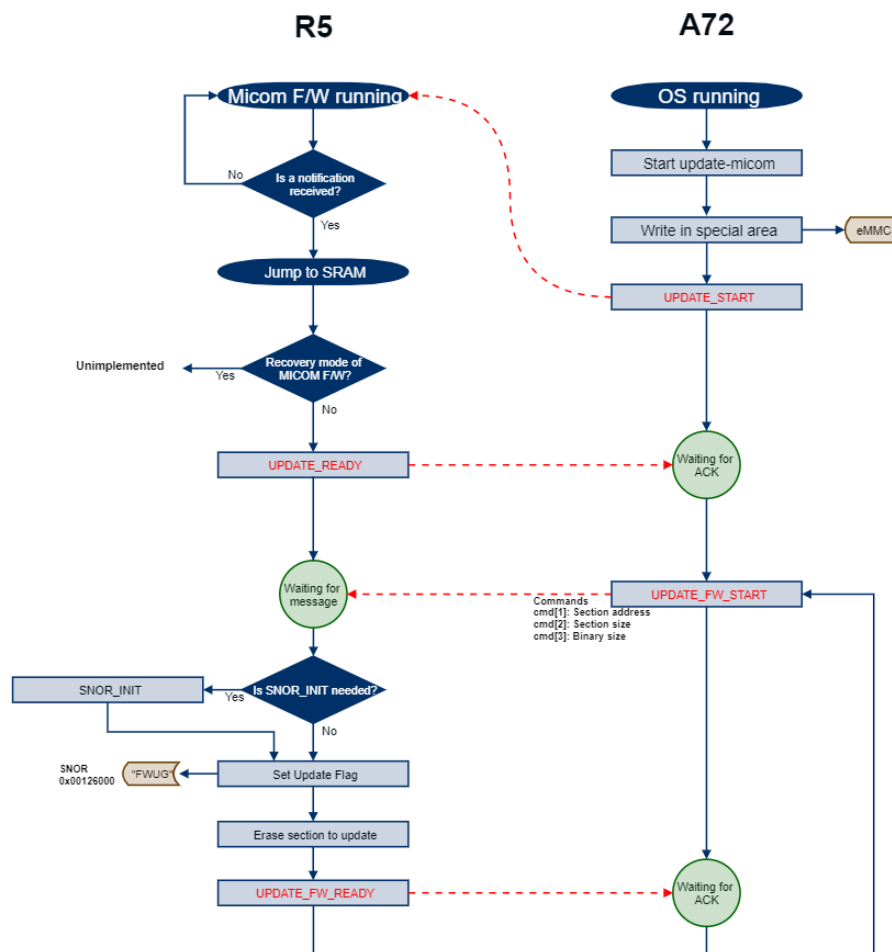


Figure 7.1 Block Diagram of FWUG Process

7.2 Operation Flow of FWUG Process

Refer to the operation flow of FWUG process.



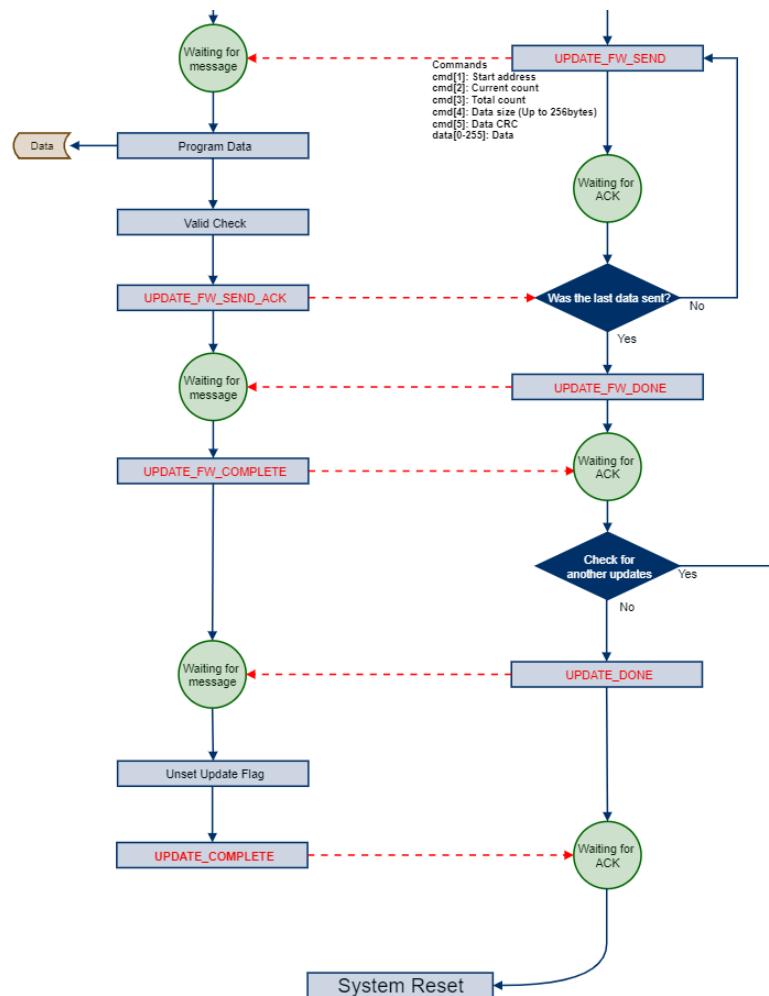


Figure 7.2 Operation Flow of FWUG Process

Refer to the following Chapter 7.2.1 – Chapter 7.2.5 for MBOX command of FWUG process.

7.2.1 UPDATE_START [ACK: UPDATE_READY]

- I. When tcc_snor_updater driver on A72 sends an UPDATE_START message to R5, the R5 executes R5 sub F/W in SRAM1 area (0x07000200).
- II. R5 checks the cause that R5 sub F/W is executed. For example, R5 sub F/W is executed when R5 main F/W binary is broken because of update or other reasons. (If necessary, implementation is required.)
- III. R5 sends ACK or NACK to A72.

7.2.2 UPDATE_FW_START [ACK: UPDATE_FW_READY]

- I. With an UPDATE_FW_START message, A72 sends R5 the start address of section to be updated and the size of section and binary to be updated.
- II. If R5 updates the first binary, R5 initializes S-NOR that is not initialized.
- III. R5 sets flag in specific S-NOR area (0x08126000) to prepare for sudden power-off or unwanted termination. If this flag is set, R5-BL1 is booted to R5 sub F/W instead of R5 main F/W.
- IV. Since S-NOR flash memory is a media that does not support overwriting, R5 erases the section, to update.
- V. R5 sends ACK or NACK to A72.

7.2.3 UPDATE_FW_SEND [ACK: UPDATE_FW_SEND_ACK]

- I. With an UPDATE_FW_SEND message, A72 sends R5 the start address to be written, current number of split data, total number of split data, split data size to be written, CRC of split data, and split data.
- II. R5 checks itself that it receives data sequentially. If it is not, R5 sends NACK to A72.
- III. R5 calculates the CRC of the data written to SNOR, and compares the CRC received from A72 via message.
- IV. R5 sends ACK or NACK to A72.
- V. A72 and R5 repeat Step I to Step IV until the binary update is completed.
- VI. If there are other binaries left to be updated, A72 and R5 repeat Step I to Step V.

7.2.4 UPDATE_FW_DONE [ACK: UPDATE_FW_COMPLETE]

- I. A72 sends an UPDATE_FW_DONE message to R5, when all of the chunk data in a binary is updated.
- II. R5 sends ACK to A72.
- III. If there is another binary that needs to be updated, A72 sends UPDATE_FW_START message to R5 and then restarts the binary update.

7.2.5 UPDATE_DONE [ACK: UPDATE_COMPLETE]

- I. When R5 receives an UPDATE_DONE message, R5 releases the update flag that is set in the UPDATE_FW_START step.
- II. R5 sends ACK to A72.
- III. A72 attempts to reset the system since S-NOR is completely updated.

7.3 How to Build R5 Sub F/W Project

R5 sub F/W is already built and included in S-NOR.

However, if you need to modify R5 sub F/W, refer to this chapter.

You can make R5 sub F/W by using R5 sub F/W project that is compressed as .zip file. You can unzip and use this project.

For more detailed information about location of R5 sub F/W project, refer to Figure 3.2.

To build the R5 sub F/W project, move to top directory of R5 sub F/W project. Then, build the project.

```
$ cd tools/FWUG
$ make tcc805x clean
$ make tcc805x
```

Figure 7.3 Make Command for R5 Sub F/W Project

Note 1: If you have a problem about build environment, check toolchain path “TOOLCHAIN_PREFIX” in “tools/FWUG/platform/tcc805x/rules.mk”.

After you build the project, a new “r5_sub_fw.rom” is made in output directory (FWUG/build-tcc805x) of R5 sub F/W project.

Then, you can replace the old “r5_sub_fw.rom” in boot-firmware to the new “r5_sub_fw.rom” in R5 sub F/W project.

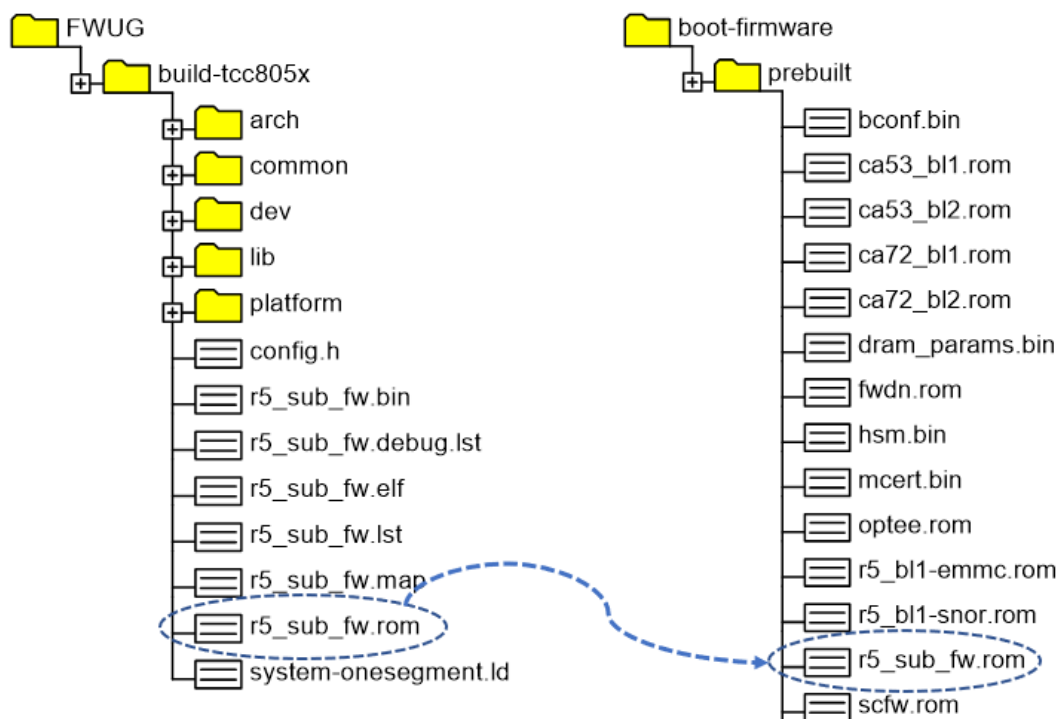


Figure 7.4 Copy New Image

You can make and update S-NOR image after replacing the “r5_sub_fw.rom”.

Note 2: Boot-firmware is not included in R5 main F/W project. For more detailed information about boot-firmware and making S-NOR image, refer to “TCC805x Linux SDK-Getting Started”. [2]

8 REFERENCES

- [1] TCC805x MCU BSP-Getting Started
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.01: 2021-01-22

- Updated
 - Chapter 5: Names of structure variables in declaration

Rev. 1.00: 2020-11-12

- First version release

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